

a) Vacancy concentration:

$$\frac{n}{N} = e^{-\frac{Q_v}{kT}}$$

$$N = \frac{N_A \rho}{A}$$

$$Q_v = 0.9 \text{ eV}$$

$$N = \frac{N_A (8.4 \times 10^6)}{63.5}$$

$$= 7.966 \times 10^{28} \text{ m}^{-3}$$

$$n = N e^{-\frac{Q_v}{kT}}$$

$$= 7.966 \times 10^{28} e^{-\frac{0.9}{k(1000+273)}}$$

$$= 2.184133449 \times 10^{25}$$

$$\approx 2.2 \times 10^5 \text{ vacancies m}^{-3}$$

$$10) \tau = 660 + 273.15$$

$$= 933.15$$

Vacancy concentration:

$$\frac{n}{N} = e^{-\frac{Q_v}{kT}}$$

$$= e^{-\frac{0.74}{8.62 \times 10^{-5} (933.15)}}$$

$$= 1.010711566 \times 10^{-4}$$

$$\approx 0.01\%$$

$$11) Q_v = 1.1 \text{ eV} \quad T = 100 + 273.15 \\ = 373.15$$

$$\frac{n}{N} = e^{-\frac{Q_v}{kT}}$$

$$\ln\left(\frac{n}{N}\right) = -\frac{Q_v}{kT}$$

$$T = -\frac{Q_v}{k \ln\left(\frac{n}{N}\right)}$$

$$\frac{s}{Z} = e^{-\frac{Q_v}{kT}}$$

$$= e^{-\frac{1.1}{8.62 \times 10^{-5} (373.15)}}$$

$$= 1.405898714 \times 10^{-5}$$

$$\frac{s}{Z} = 1.41 \times 10^{-5} (1.5)$$

$$= 2.108848071 \times 10^{-15}$$

$$T = \frac{-1.1}{8.62 \times 10^{-5} \ln(2.11 \times 10^{-15})}$$

$$= 377.6272865 \text{ K}$$

$$\approx 377.63 \text{ K}$$

12)

1. Heat to a higher temperature, hold for sufficient time so that the higher vacancy concentration at that temperature is attained. Then, rapidly quench to room temperature.

Due to the rapid quenching process, the vacancies do not have enough time to anneal out.

Hence, sample B can have a higher concentration than sample A.

2. Irradiation of sample B with high energy particles. This increases vacancy concentration and also increases self-interstitials.

3. Severe plastic deformation is another possibility. Dislocation can also increase substantially.